



Tribhuvan University
Faculty of Humanities and Social Science

A
PROJECT PROPOSAL
ON
INTELLIGENT CAREER RECOMMENDATION SYSTEM

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Department of Computer Application
Sagarmatha College of Science and Technology
Sanepa, Lalitpur, Nepal

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Submitted By

Prajwol Rajbansi

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ABSTRACT

The proposed **Intelligent Career Recommendation System Using Machine Learning** is designed to help students make informed career decisions by analyzing their skills, interests, academic background, and personal preferences. In today's rapidly evolving technological environment, students often struggle to identify the most suitable career path due to the increasing number of professional roles and changing industry demands. Traditional career guidance methods are often broad, less personalized, and may not reflect current employment trends.

This project proposes a web-based intelligent recommendation platform that uses machine learning techniques to suggest suitable career options, especially for students in the computer and information technology domain. The system will collect user input, preprocess it, and pass it to a trained machine learning classifier that predicts the most relevant career category. In addition to recommending possible professions, the system will also provide supporting information regarding required skills and possible areas for improvement.

The main aim of this project is to develop a simple, practical, and user-friendly prototype that reduces confusion in career planning and helps students better understand their future opportunities. The expected result is a functional recommendation system that can support students, academic institutions, and counselors in providing smarter and more personalized career guidance.

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Chapter 1

INTRODUCTION

1.1 Background

In the modern age, selecting the right career path has become a difficult task for many students. Rapid technological advancement, changing job market trends, and the continuous emergence of new professional roles have made career planning more complicated than ever before. Students often have limited awareness of available opportunities and may not know which profession best matches their interests, strengths, and academic preparation.

In the field of computer applications and information technology, students can choose from many career options such as software development, web development, data analysis, cybersecurity, cloud computing, network engineering, and artificial intelligence. Although these opportunities are valuable, they also create confusion among students who are trying to determine the most appropriate direction for their future.

To address this challenge, this project proposes the development of an **Intelligent Career Recommendation System Using Machine Learning**. The system will use machine learning techniques to analyze student profiles and recommend suitable career options based on the user's skills, interests, and academic background. This will support students in making more confident and informed career decisions.

1.2 Motivation

The motivation behind this project comes from the need for a smarter and more personalized career guidance solution. Traditional career counseling often provides general advice that may not fully reflect the specific abilities, interests, and future goals of each student.

Many students commonly face questions such as:

- Which IT career is best suited for my profile?
- What skills are required for a specific profession?
- Which area matches me more: development, networking, security, or analytics?
- How can I align my interests with current market demand?

This project is motivated by the need to answer such questions through an intelligent recommendation platform that can provide meaningful and data-driven guidance.

Chapter 2

PROBLEM STATEMENT

The development of the **Intelligent Career Recommendation System Using Machine Learning** is proposed to solve the following key problems observed among students:

- **Lack of Personalized Guidance:** Many students do not receive career advice that is tailored to their own profile, interests, and technical abilities.
- **Limited Awareness of Career Options:** Students are often unaware of emerging and specialized roles in the IT and computer application field.
- **Skill-Career Mismatch:** Students may possess useful skills but may not know which career options align with those skills.
- **Changing Job Market Trends:** The technology job market changes rapidly, making it difficult to depend only on traditional and static guidance methods.
- **Confusion in Career Planning:** Due to lack of proper guidance, students may choose unsuitable professions or fail to prepare for better opportunities.

Therefore, there is a need for a practical and intelligent system that can analyze student information and recommend relevant career paths in a simple and accessible way.

Chapter 3

OBJECTIVES

The main objectives of the proposed system are as follows:

- To develop a web-based intelligent career recommendation system using machine learning.
- To recommend suitable career options based on user skills, interests, and academic background.
- To help students understand which profession best matches their profile.
- To identify skill gaps related to recommended careers.
- To provide a simple and user-friendly platform for career guidance.

Chapter 4

METHODOLOGY

4.1 System Architecture

The proposed system follows a workflow consisting of user input, preprocessing, model prediction, and recommendation output.

1. The user enters information such as skills, interests, educational background, and career preferences.
2. The system preprocesses the input data into a structured feature vector.
3. A trained machine learning classifier analyzes the input.
4. The system predicts and displays the most suitable career option.
5. Supporting details about the recommended career and related skills are shown to the user.

4.2 Machine Learning Approach

The proposed recommendation engine will use a **supervised multiclass classification** approach. A dataset containing student-related features and corresponding career labels will be prepared first. Input features such as programming interests, technical skills, preferred domain, academic strengths, and career preferences will be encoded into machine-readable form.

The dataset will then be divided into training and testing sets. Multiple classification algorithms such as **Decision Tree**, **Random Forest**, and **Logistic Regression** will be tested. The model with better performance will be selected as the final recommendation engine. Model evaluation will be done using measures such as **accuracy**, **classification report**, and **confusion matrix**. After training, the final model will predict one of several career classes, such as Web Developer, Data Analyst, Software Developer, Network Engineer, or Cybersecurity Analyst.

4.3 Frontend

The frontend of the system will provide an interactive and user-friendly interface where users can input their data and receive career recommendations. It may be developed using:

- HTML
- CSS
- JavaScript
- Streamlit or similar web-based interface tools

4.4 Backend

The backend will process the input data and generate recommendations. It may be developed using:

- Python
- Scikit-learn
- Pandas
- NumPy

4.5 Technology Stack

The proposed technology stack includes:

1. **Frontend:** HTML, CSS, JavaScript / Streamlit
2. **Backend:** Python
3. **Libraries:** Pandas, NumPy, Scikit-learn
4. **Dataset Format:** CSV or other structured dataset

Chapter 5

TIME LIMIT

5.1 Gantt Chart

The following table illustrates the proposed timeline for the execution of the project over a span of eight development weeks:

SN	Project Activities	Development Weeks							
		1	2	3	4	5	6	7	8
1	Planning								
2	System Analysis								
3	System Design								
4	Data Collection and Preprocessing								
5	Model Development and Training								
6	Testing and Evaluation								
7	Deployment and Documentation								

Table 5.1: Gantt Chart Output

Chapter 6

EXPECTED OUTCOME

At the completion of this project, the expected outcomes are:

1. A functional web-based career recommendation system.
2. Personalized career suggestions for users.
3. Better understanding of suitable IT and computer-related career paths.
4. Improved awareness of the skills required for different professions.
5. A practical prototype that can be further improved in the future.

Ultimately, the project is expected to help students reduce confusion and make more confident career decisions with the support of machine learning.

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